

# Yuriy Korotovskikh

SENIOR FULLSTACK DEVELOPER · WEB3 ENGINEER

Yerevan, Republic of Armenia

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## Summary

Senior Fullstack / Web3 Engineer building wallet, payments, and dApp flows since 2020.

I work where product UX meets blockchain complexity: wallet connection, signing, top-ups, transaction states, SDKs, indexers, and backend infrastructure.

Built crypto payment products, dApps from scratch, explorers, trading interfaces, smart contract tools, SDKs, and reusable UI systems across Web3 and fintech.

Shipped products such as smart contracts playground, block explorers, CLIs, several production dApps, taking them from early-stage ideas to reliable releases with a focus on usability and delivery.

## Skills

|                  |                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b> | <b>Proficiency</b> - JavaScript, TypeScript, Solidity, HTML, CSS, <b>Working Knowledge</b> - Go, Elixir, Python, Rust                                 |
| <b>Frontend</b>  | React, Next.js, Astro, Vue.js, Redux, Effector, Zustand, Web Workers, Storybook, Vite, Rollup, Tailwind, WebAssembly, TanStack, Frontend architecture |
| <b>Web3</b>      | Viem, Wagmi, Ethers, MetaMask, WalletConnect, Ledger, RainbowKit, Reown AppKit                                                                        |
| <b>DevOps</b>    | Docker, Ansible, TeamCity, Nix, GitHub Actions CI                                                                                                     |
| <b>Backend</b>   | Node.js, SQL, REST, JSON-RPC, WebSocket                                                                                                               |
| <b>Testing</b>   | Jest, Vitest, Cypress, Playwright, Synpress, Hardhat                                                                                                  |
| <b>UX</b>        | Interaction design, Figma, Design systems, Accessibility                                                                                              |

## Work Experience

### Mercuryo

London, United Kingdom (remote)

SENIOR FRONTEND DEVELOPER

Sept. 2025 - Current

*Mercuryo is a crypto-powered payments infrastructure provider whose core products include a wallet-integrated on/off-ramp widget, Spend Card, and Wallet (10M users as of May 2025).*

- Owned the Mastercard Crypto Credentials frontend, including wallet linking, message and transaction signing, and error-handling flows across partner wallets. Led the migration from Web3Modal to Reown AppKit.
- Delivered end-to-end frontend flows for Spend Card and Ledger Spend users: eligibility checks, multi-level KYC flows, Sumsb integration, card issuance, top-ups, limits, transaction history, balance states, and UI mocks.
- Integrated Apple Pay, Google Pay, and Cloudflare Turnstile into Mercuryo frontend flows.
- Built and maintained partner wallet integrations (Ledger, MetaMask, Trust Wallet, and others) in a React + Redux widget with white-label embedding, theming, advanced customization, localization, and iframe-host communication.
- Integrated core Web3 UX primitives: WalletConnect/Reown AppKit connection, multi-chain EVM support, message and transaction signing across wallets.
- Expanded automated test coverage for high-risk payment flows using Playwright E2E and Vitest unit tests (on/off-ramp, KYC redirects, approvals and signatures, and card onboarding), and built a sharded Playwright pipeline on self-hosted runners with Allure reporting.
- Led frontend observability, incident response, security and code quality initiatives, including Sentry instrumentation, product analytics, Snyk monitoring, dependency management, static analysis tooling (zizmor, React Doctor, ESLint, Oxc), and nightly CI checks.
- Shaped frontend architecture with FSD-inspired module boundaries, separation of concerns, dependency flow rules, and code quality gates to keep a large white-label codebase maintainable. Contributed to frontend engineering standards, reusable component systems, and scalable development practices across the team.
- Built AI-assisted engineering workflows by authoring multi-level agent guides, harness tooling, and reusable internal skills for code generation, test authoring, and review checklists.
- Led frontend onboarding and learning initiatives through internal workshops and documentation, and regular updates on frontend trends and best practices.

### Diffuse Prime

Limassol, Cyprus (remote, contract)

FOUNDING FRONTEND ENGINEER

Aug. 2025 - Mar. 2026

*Diffuse Prime is a ZK/TEE-powered non-custodial lending protocol for structured leverage deals.*

- Designed architecture and built a Next.js + TypeScript frontend dApp from scratch, including wallet connection, on-chain interactions, user flows for lenders and borrowers. Integrated Sentry, logging, and event analytics to monitor the product.
- In collaboration with the design team, implemented a modern ESM UI kit with Radix UI, Tailwind, and TanStack libraries.
- Implemented a testing strategy with Vitest unit tests, visual regression tests, Lighthouse tests, and Playwright end-to-end tests with local network fixtures. Covered more than 90% of the codebase with tests.

- Designed and implemented full cycle CI/CD pipelines for the frontend, ensuring integration and delivery to Vercel.
- Designed and implemented a Node.js indexer for processing on-chain events and integrated it with the Next.js API and Vercel cron jobs.
- Implemented a custom JavaScript SDK for interacting with the protocol, which was used across the frontend and other tools.
- Organized monorepo with npm workspaces, implemented automated release and versioning process with changesets, introduced best security practices and audits, and set up GitHub Actions CI.
- Collaborated closely with founders as a sole frontend engineer to design and implement various features and visual elements, to ensure the product meets the needs of its users.
- Designed repository AI workflows by authoring agent operating rules, building task-specific agents (feature delivery, refactoring), and creating reusable prompt patterns for consistent output quality.

## Nil foundation

Limassol, Cyprus (remote)

FULL-STACK DEVELOPER

May. 2022 - Aug. 2025

*Nil Foundation is an Ethereum L2 startup working on Ethereum horizontal scalability.*

- Built from scratch and released proof market DEX-like interface, which is a decentralized exchange of computation powers for proofs generation.
- Built a block explorer for the NIL Layer 2 chain with detailed views for blocks, transactions, and accounts. Designed and optimized the frontend architecture for performance and scalability. Also contributed to the Node.js backend powering the explorer.
- Developed an interactive, sharding-specific smart contract playground for the NIL chain. The tool enables users to write and compile Solidity code, save snippets, deploy contracts, and interact with them — all within a unified interface.
- Built a custom JavaScript SDK (web and Node.js) for interacting with the Nil blockchain, tailored to support SSZ serialization and Poseidon hashing. This library became the foundation for multiple critical tools, including a CLI, Hardhat plugin, and browser wallet extension.
- Developed and maintained the corporate website using Next.js with a Strapi CMS backend, along with additional marketing pages to support product outreach.
- Created and maintained a reusable React UI component library, adopted across nearly all NIL interfaces to ensure design consistency and development efficiency.
- Built and maintained CI/CD pipelines using GitHub Actions CI, Ansible, and Jinja for frontend and backend projects.

## InfoTeCS JSC

Saint Petersburg, Russia (on-site)

FRONTEND DEVELOPER

Mar. 2021 - May. 2022

*InfoTeCS is a cybersecurity company providing a range of secure communication solutions for corporate clients.*

- Built Remote Work Management System, a big internal React app, where employees could request remote work, track approval pipelines, and submit daily reports. I created interfaces, pages and forms, worked on performance optimizations and managed API workflows with Redux Saga. I also contributed to backend C# controllers and validations to support frontend logic.
- Contributed to Employee performance evaluation system - a large-scale web application built with ASP.NET and a legacy frontend stack using jQuery and Backbone.js. Worked on API, controllers and database integrations.
- Developed React-based Artifact Management App. This internal tool allowed teams to upload, download, and manage build artifacts across multiple systems. All existing artifacts were aggregated in a dynamic responsive table with server-side sorting and filtering, making it easy for engineers to quickly locate, and interact with the files.
- Led frontend development for a Next.js documentation portal targeting foreign branches, combining static site generation with dynamic internal routing. This portal was a key touchpoint between the company and its foreign customers. One part of the app was public-facing, the other integrated into an internal doc system.
- Maintained React components library and closely collaborated with Design department to ensure consistent implementation of the design system.

## Oday Technologies

Moscow, Russia (remote)

FRONTEND DEVELOPER

Sept. 2020 - Mar. 2021

*Oday Technologies is a company specializing in the development of automation tools and information security.*

- Designed and developed a Vue 2 and Vuetify-based monitoring service, integrating interactive Leaflet maps for real-time data visualization and location tracking.
- Worked on a financial analytics React application with a Python backend, where I developed UI elements and collaborated with backend team to implement advanced search across multiple sources.

## Education

### Kurgan State University (KSU)

Kurgan, Russia

BACHELOR'S DEGREE

Sep. 2014 - Aug. 2018

### Saint Petersburg State University of Economics (UNECON)

Saint Petersburg, Russia

MASTER'S DEGREE

Sep. 2018 - Jul. 2020